

Shubham Kumar

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Professional Summary

Software Engineer building scalable full-stack applications, AI-powered platforms, and real-time systems with a focus on performance, reliability, and clean architecture.

Technical Skills

Languages: JavaScript, TypeScript, Python, C++, SQL
Frontend: React.js, Next.js, Tailwind CSS, Redux Toolkit, Context API, Framer Motion, GSAP
Backend: Node.js, Express.js, REST APIs, WebSockets, SSE, MQTT, PostgreSQL, JWT, OAuth
Cloud: AWS EC2, Docker, Vercel
Tools: Git, WebContainers, Postman
AI: OpenAI API, LLM Orchestration, Prompt Engineering

Work Experience

SnetLabs Dec 2024 – Present
Software Engineer Noida, UP

- Co-architected a browser-side control plane for an enterprise VMS using a custom playback engine driven by rAF (requestAnimationFrame), enabling frame-accurate sync across 100+ concurrent camera streams.
- Replaced polling with a WebSocket and MQTT signalling layer, achieving sub-50ms latency for PTZ camera controls and instant AI alert delivery — eliminating a core reliability bottleneck in the platform.
- Implemented Viewport Virtualisation with scroll-throttled subscribe/unsubscribe signals, cutting stream signalling load proportionally to off-screen feeds and keeping the UI fluid under heavy grid layouts.
- Partitioned React Context and batched state updates to reduce re-renders by 40–50%, sustaining a consistent 60 FPS experience across high-density, data-heavy dashboard views.
- Built a React Canvas visual workflow editor for composing AI analytics pipelines graphically, enabling non-technical operators to configure and deploy complex worker processes without engineering support.
- Co-authored **labsense-ui-kit** (npm) — a shared React + TypeScript component library that standardised design patterns across teams and accelerated feature delivery by 25%.

Potterstech June 2023 – Aug 2023
Frontend Engineer Intern Remote

- Built a library of responsive, reusable ReactJS + Tailwind CSS components for a cross-browser production application, shipped to paying customers within the internship window.
- Integrated RESTful API endpoints with full async lifecycle handling — loading skeletons, error boundaries, and empty states — significantly improving perceived performance and UX consistency.
- Implemented interactive UI features using React Hooks and modern JS patterns, reducing boilerplate and improving render efficiency across feature modules.

Featured Projects

JustSay – AI Website Builder TypeScript, React, Node, PostgreSQL

- Built a production SaaS that converts natural language prompts into fully functional Vite + React apps, executed live inside WebContainers — zero server-side runtime, zero file system access, and instant in-browser preview.
- Engineered a 7-stage LLM orchestration pipeline (classify → expand → plan → skeleton → batch-generate → normalise → fix), streaming generated code token-by-token via SSE for real-time file materialisation.
- Designed a two-tier caching system using exact prompt fingerprints and semantic category matching, reducing redundant LLM inference, API costs, and generation latency.
- Architecting multi-tenant WebContainer lifecycle management with resource quotas, idle detection, and graceful teardown to support concurrent user workloads efficiently.

WeTube – Full-Stack Video Platform MERN, Firebase

- Architected a full-stack YouTube-style platform with Express and MongoDB powering video management, subscriptions, comments, likes, playlists, and channel workflows.
- Integrated Firebase Storage for secure video and thumbnail uploads with progress tracking, protected using JWT and Google OAuth authentication.
- Implemented normalized Redux Toolkit state management to eliminate redundant API requests and improve client-side performance and responsiveness.
- Deployed the application on Vercel and Render with environment-based configuration, responsive design, and optimized asset delivery.

JobMailer – Intelligent Email Campaign Platform

React, Node, Nodemailer

- Built a full-stack email campaign platform that automates personalized job applications by processing recruiter contact sheets, rendering dynamic templates, and attaching resumes at scale.
- Designed a queue-based email delivery engine with campaign state management, retry handling, rate limiting, and randomized dispatch intervals to improve reliability and reduce spam detection.
- Integrated OAuth 2.0, Google Sheets synchronization, advanced analytics, and AI-assisted email generation to enable secure, large-scale recruitment outreach campaigns.

TSP via Differential Evolution

Python, NumPy, Matplotlib

- Designed a Differential Evolution solver for the Travelling Salesman Problem using custom mutation (DE/rand/1) and crossover strategies to optimize route discovery.
- Outperformed Genetic Algorithm baselines in convergence speed and solution quality across multiple benchmark datasets.
- Implemented adaptive crossover tuning and population diversity preservation to prevent premature convergence on complex optimization problems.

Education

University of Petroleum and Energy Studies
B.Tech in Computer Science

2020 – 2024
Dehradun, Uttarakhand